

Topic: Basic Instruction Cycle with Interrupt Processing

Q. Describe the phases of a Basic Instruction Cycle. Explain how the CPU handles interrupts by modifying the standard instruction cycle.

[DEF] Definition to Remember

The Instruction Cycle is the fundamental operational process of a CPU, consisting of three phases: Fetch, Decode, Execute. When an external event demands immediate attention, the CPU modifies this cycle by appending an Interrupt Cycle to save state and run an Interrupt Service Routine (ISR).

Basic Instruction Cycle & Interrupt Servicing

1. Fetch Cycle:

* CP

2. Handling Interrupts

An Interrupt breaks the normal sequential flow. To handle it, the CPU adds an Interrupt Cycle immediately after the Execute cycle.

Interrupt Processing Steps:

1. Ch

[MUST-WRITE] Must-Write Points (for 10 marks)

1. Standard cycle: Fetch -> Decode -> Execute.
2. Fetch: Gets instruction via PC, stores in IR, increments PC.
3. Decode: Control unit translates the Opcode.
4. Execute: ALU performs the task.
5. Interrupt Cycle occurs at the end of the Execute phase.
6. CPU suspends program and saves the Program Counter (PC) and Flags to the Stack.
7. CPU executes the Interrupt Service Routine (ISR), then uses IRET to pop the PC and resume the original program.

[RECALL] Quick Recall

Cycle: Fetch (PC to IR) -> Decode (CU) -> Execute (ALU) -> Check Interrupt -> Save PC to Stack -> Run ISR -> IRET (Pop PC) -> Resume